Notes on the Graph Minor Theorem

The Graph Minor Theorem is, by all accounts, one of the most important theories in Graph theory. It proved a long open conjecture of Wagner, and has countless application, but as much much as the theorem itself, the theory developed in the twenty papers that Roberston and Seymour wrote to prove it have had an immense impact on the landscape of structural Graph Theory. We start working from Chap 12 of Diestel, which is, or at least was at the time of its writing in 2005, widely considered THE introduction to the Theorem.

1 Introduction

Its all about minors, so lets start with a definition. (This is from Chaper 2.1 of Diestel).

Definition 1.1. An M-minor of a graph G is a map $V : V(H) \to 2^{V(G)} : h \mapsto V_h$ such that the *branch sets* V_h are disjoint, each induces a connected subgraph of G, and $h \sim h'$ in H if and only if there are $v \in V_h$ and $v' \in V_{h'}$ with $v \sim v'$ in G.

Alternately, an M-minor of G is the subgraph of G induced by the image of the map V. It is clear that by contracting the edges inside each branch set, we get a copy of M, so a graph G has an M-minor if and only if we can get M from G by removing and contracting edges.

If G has an M minor, we write $M \preccurlyeq G$. It is easy to see that the minor relation '$\preccurlyeq$' is a partial order on the set of isomorphism classes of finite graphs. (If we we consider labelled graphs, isomorphic but distinct graphs can be minors of each other, and two non-isomorphic infinite graphs can me minors of each oher, so it is often called a *quasi order* rather than a partial order.)

Recall that Kurotowski showed that a graph is planar if and only if it has no $K_{3,3}$ or K_5 minor. We started this proof with the easy observation that if a graph has a non-planar minor then the graph is non-planar. A fancy way of saying this is that the class of planar graphs is *minor closed*: any minor of any graph in the class is also in the class.

Any minor closed class $\mathcal{C}$ of graphs, can be described by the set $\mathcal{X}$ of graphs that none of its graphs contain as a minor. Indeed we write

$$\mathcal{C} = \mathrm{Forb}_{\preccurlyeq}(\mathcal{X})$$

and call $\mathcal{X}$ its set of *excluded*, (or *forbidden*, explaining the notation,) minors. We may always restrict $\mathcal{X}$ to its anti-chain of $\preccurlyeq$-minimal elements, as excluding a minor excludes any minor above it.

It is easy to see that the class of graphs embeddable on any given surface is a minor closed class, so is defined, as planar graphs are in Kurotowski's theorem, by a set of excluded minors.

This is all trivial, and is a prime example of where the Graph Minor Theorem has application. The Graph Minor Theorem says that the minor partial order does not have infinite anti-chains. So any minor closed class is defined by a **finite** set of excluded minors.

Another result connected to the Graph Minor Theorem is that for any minor M there is a polynomial time algorithm to check if a given graph has M as a minor. It follows that we can check in polynomial time if the graph is in a given minor closed class.

While we say 'finite' and 'polynomial', the constants involved in these statements are huge, and there is very active work in getting these theoretically polynomial algorithms to be more practical. Hopefully we can start to understand some of these things.

2 Diestel on the Graph Minor Theorem

2.1 The Minor Order

A quasi order, or partial order, is *well* ordered if there it has no infinite antichains or infinite descending chains. The minor order of finite graphs can have no infinite descending chains, so that it is well ordered means that it has no infinite anti-chains. This means that any infinite sequence of graphs contains a infinite increasing chain in the minor order.

The Graph Minor Theorem of Roberson and Seymour shows that the minor order has no infinite anti-chains.

Theorem 2.1. *The minor order of finite graphs is a well quasi order.*

In this section Diestel proves the following lemma.

Lemma 2.2. *If X is well quasi ordered, then so is the family of finite subsets of X under the induced ordering defined, for $A, B \subset X$, by $A \leq B$ if there is an injection $f : A \to B$ such that for every $a \in B$, $a \leq f(b)$.*

The proof uses the following useful equivalent definition of a well ordered set. For an infinite sequence $(a_n)_{n\in\mathbb{N}}$ of the ordered set, a pair is *good* if there is some $i < j$ such that $a_i \leq a_j$. A partial order is well ordered if and only if every infinite sequence contains a good pair. We will use this notion below.

2.2 The Graph Minor Theorem for Trees

Topological minors are introduced in Section 1.7 of Diestel, they are a bit less general than minors, but they play a similar role. Kuratowski actually showed that a graph is planar if and only if it excludes K_5 and $K_{3,3}$ as topological minors. The version we use more often now, with minors, is actually due to Wagner.

A *subdivision* of a graph T is a graph we get from it by subdividing some of its edges into paths. A graph G has T as a topological minor if it contains some subdivision of T as a subgraph.

We show the following

Theorem 2.3 (Kruskal (1960))**.** *The class of finite trees under the topological minor order is well ordered.*

Proof. Consider rather the set of rooted trees (digraph trees oriented towards the root) and say that for two such trees T and T', $T \leq T'$ if some subdivision of T is isomorphic to some subgraph of T'. We have that $T \leq T'$ implies that the tree T (with orientation removed) is a topological minor of the tree T'. The set of rooted trees contains the set of trees and $\leq$ restricts on them to the topological minor order, so proving that $\leq$ is well ordered is stronger than proving that the topological minor order of trees is.

We show that $\leq$ is well ordered by contradiction. Towards contradiction, assume that there is some bad sequence– a infinite sequence of rooted trees with no good pair. Of these bad sequences, choose one bad sequence $(T_n)_{n\in\mathbb{N}}$ that lexicographically minimises the sequence $(|T_n|)_{n\in\mathbb{N}}$ of orders. Let A_n be the set of components (rooted trees) left when we remove the root from T_n.

Claim 2.4. *The set $A = \bigcup_{n\in\mathbb{N}} A_n$ is well ordered.*

Proof. Let $(S_n)_{n\in\mathbb{N}}$ be an infinite sequence of trees of A, and for each n let $f(n)$ be such that $S_n \in A_{f(n)}$ and let k be such that $f(k)$ is the minimum of the $f(n)$. Our goal is to show that (S_n) is good. As $S_k \in A_{f(k)}$ we have that $S_k \leq T_{f(k)}$ and so by choice of the the sequence $(T_n)_{n\in\mathbb{N}}$, the sequence

$$T_0, T_1, \ldots, T_{f(k)-1}, S_k, S_{k+1}, \ldots$$

is good; and so has a good pair $T \leq T'$. Not both of T and T' are in $T_0, \ldots, T_{f(k)-1}$, as this is a bad sequence; and if they are both in $S_k, S_{k+1}, \ldots$, then (S_n) is good, as needed, so we may assume that T is T_i for $i \leq f(k) - 1$ and T' is S_j for $j \geq f(k)$. But then T' is in $A_{f(j)}$ so $T' \leq T_{f(j)}$, and we have $i \leq f(k-1) < f(k) \leq f(j)$, showing that $(T_i, T_{f(j)})$ is a good pair; this however, is impossible, as (T_n) is bad. $\diamond$

As A is well ordered, the Lemma tells us that the set of finite subsets of A is well ordered, and so the sequence $(A_n)_{n\in\mathbb{Z}^*}$ must have a good pair $A_i \leq A_j$ from which it is clear (with the right picture) that $T_i \leq T_j$, giving a contradiction.

□

2.3 Tree Decompositions

Definition 2.5. A *tree decomposition* (T, V) of a graph G consists of a tree T, and a function $V : V(T) \to 2^{V(G)}$ that takes a vertex i of T to a set $V_i \subset V(G)$ of vertices of G called a *bag*, such that every edge of G has both endpoints in some bag, and the *coherence* property is satisfied: if the vertex j on the path between i and k in T, then

$$V_j \subset V_i \cap V_k.$$

Problem 2.1. Show that the following is an alternate statement of coherence: For any vertex v of G the set of bags that contain a vertex v induces a subtree of T.

The *width* of a tree-decompostion (T, V) of a graph G is one less than the size of the largest bag. The *tree width* $\mathrm{tw}(G)$ of a graph G is the min of the widths over all tree decompositions.

A graph has tree width 1 if and only if it is a forest.

Problem 2.2. Show that a tree has tree width 1, that a cycle has tree width 2, and that an n-clique has tree width $n-1$.

One can draw a tree-like picture of this decompostion. From this picture, the following separations facts, proved in Diestal, are eminently believable.

The following results are about how separations in the tree T yield separations in G. Recall that a set $S \subset V(G)$ is a *separator* of G, or *separates* G, if $G \backslash S$ is disconnected; a pair of subgraphs U and U' such that $V(U) \cup V(U') = V(G)$ and $V(U) \cap V(U')$ separates G is a *separation*.

Lemma 2.6 (Lemma 12.3.1 of Diestal)**.** *For any edge ij of T, where T_i and T_j are the components of $T \setminus ij$, the set $V_i \cap V_j$ is a separator of G into the separation $(U_i.U_j)$ where $U_i = \bigcup_{v\in T_i} V_v$ and $U_j = \bigcup_{v\in T_j} V_v$.*

Here is a stronger statement than that that K_n has tree width $n-1$. The proof is one that is used again.

Fact 2.7. *For any tree decomposition (T, V) of a graph G, any clique K of G is in a bag V_i.*

Proof. Assume not. Then K certainly cannot be in $C := V_i \cap V_j$ for any edge ij of T, and there can only be vertices in one side, U_i or U_j of the separation. Orient the edge ij of T towards the side with a vertex of K. Any vertex of T has at most one out edge, and so there is a sink s. So K is in V_s. □

The following version of Lemma 2.6 is also useful.

Lemma 2.8 (Lemma 12.3.4 of Diestal)**.** *For any set of vertices W of G not in a single bag of (T, B), there is an edge tt' of T such that $V_t \cap V_{t'}$ separates W.*

In particular, the tree width of a tree decomposition of G upper bounds the connectivity of T. A k-connected subgraph in G pushes the tree width of G up past k.

Now we look at how tree decompositions persist to substructures.

Lemma 2.9 (Lemma 12.3.2 of Diestal)**.** *Taking any subgraph H of G, we get a tree decomposition of H with the same tree T by intersecting bags with $V(H)$. (Bags can be empty.)*

The following minor version of the previous result is easy to prove. So tree decompositions are 'maintained' under taking subgraphs and taking minors.

Lemma 2.10 (Lemma 12.3.3 of Diestal)**.** *Lemma 12.3.3 of Diestal. If G has tree-decomposition (T, B) and M is a minor with branch set function V, then H has a tree decomposition (T, W) in which the bag W_i consists of the vertices h of H such that V_h that contains a vertex from the bag B_i.*

From Lemma 2.10 we have that $M \preccurlyeq G$ then $\mathrm{tw}(M) \leq \mathrm{tw}(G)$.

Tree width is one of those ideas that appears everywhere now. It a class of graphs has bounded tree width, then in a lot of ways the class is similar to the class of trees. We know, for example, that any CSP problem can be solved in polynomial time for graphs of bounded tree width. The rough idea is that we start with a leaf bag, and find all possible solutions to the problem on that bag. As the size of the bag is bounded by a constant, we can do this is some constant time. We only have to keep the restriction of these solutions on the intersection of that bag with the neighbouring bags, and then find solutions on these bags that agree on the intersection. As we proceed along the tree, we only have to keep the solutions on the intersections with the unprocessed bags.

For a more explicit example of how trees of bounded tree width are tree like, Diestal give the following result of Roberson and Seymour, which takes take Theorem 2.3 a step further in the direction of the Graph Minor Theorem. He says that the proof iterates the proof of Thereom 2.3, but he doesn't give it. Presumably it is quite long.

Theorem 2.11. *For every integer $k \geq 0$, then graphs of tree-width at most k are well quasi ordered under the minor relation.*

This specialized version of the Graph Minor Theorem does not deal with graphs of large tree width, so we need to know something about such graphs. We have seen that highly connected subgraphs give large tree width, but these can take many forms. We will see several useful forms. One will be a large

grid minor. The the first version that we look at is a bramble. A *bramble* in a graph G is a set $(S_1, \ldots, S_b)$ of subsets of $V(G)$ (bramble-sets), inducing connected subgraphs, such any two touch: either share a vertex or have an edge between them. A *cover* of a bramble in G is a set $C \subset V(G)$ that has non-empty intersection with each set S_i of the bramble. The size of the smallest cover C is the *order* of the bramble.

Example 2.12. A k-clique in which each vertex is a set of the bramble is clearly a bramble of order k.

Example 2.13. A $k \times k$ grid is a bramble: for sets take the k^2 'crosses' $R_i \cup C_j$ for each row R_i and column C_j. This bramble has order k: any row or column is a cover of size k, and any smaller subset misses at least one row and one column, so misses a cross.

This proof in Diestal is about two pages. We should all try to read and understand it.

Theorem 2.14 (Seymour and Thomas (1993))**.** *A graph G has tree-width $\mathrm{tw}(G) \geq k$ if and only if it has a bramble of order greater than k.*

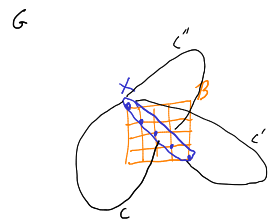
Proof. One direction is easy. We are done if some bag V_i covers $\mathcal{B}$, so we may assume that for any V_i there is some $B \in \mathcal{B}$ that is disjoint from V_i. For each edge ij of T, there is therefore some B completly in one of the sides U_i or U_j of the separator $V_i \cap V_j$, orient ij towards the side that contains some B. (No B' can be in the other side, as bramble-sets touch.) This gives an orientation of T in which any vertex has at most one out arc, so T has a sink s. Thus V_s contains B, and no vertex of B is in another bag. As B touches all other bramble-sets, V_s covers $\mathcal{B}$, so has width at least k.

Assumptions are blue. Objectives are red.

The other direction is harder. We assume that G has no brambles of order greater than k, and show that G has tree width less than k.

For a given bramble $\mathcal{B}$ of G, a tree decomposition (T, V) is $\mathcal{B}$-admissible if no bag of size $\geq k$ covers $\mathcal{B}$. We show for every bramble $\mathcal{B}$ of G that G has a $\mathcal{B}$-admissible tree decomposition. Taking the empty bramble $\mathcal{B} = 0$, every set covers it, so this implies that the $\mathcal{B}$-admissible tree decomposition has width $< k$. In proving that G has a $\mathcal{B}$-admissible tree decomposition, we assume that G has a $\mathcal{B}'$-admissible tree decomposition for any bramble $\mathcal{B}'$ with more parts than $\mathcal{B}$. We can do so because there is a bramble with maximum size– no bramble can have more than $2^{|G|}$ sets.

So fix $\mathcal{B}$, we show that G has a $\mathcal{B}$-admissible tree decomposition. We assume that for any bramble $\mathcal{B}'$ that is larger than $\mathcal{B}$ there is a $\mathcal{B}'$-admissible tree decomposition of G. Let X be a minimum cover of $\mathcal{B}$. As we assume that G has no brambles of order greater than k, we have $\ell := |X| \leq k$.



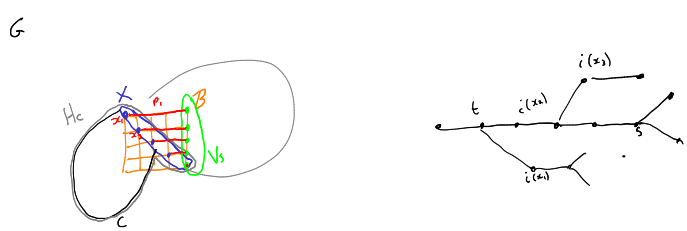
If we can show that for every component C of $G \setminus X$ there is a $\mathcal{B}$-admissible tree decomposition of $H_C := G[C \cup X]$ with X as a bag, then we can glue these decompositions together on the bag X and get a $\mathcal{B}$-admissible tree decompositon of G. So we fix a component C of $G \setminus X$ and show that H_C has a $\mathcal{B}$-admissible tree decomposition.

The easy case is that $\mathcal{B}' := \mathcal{B} \cup \{C\}$ is not a bramble. In this case C does not touch some bramble-set B of $\mathcal{B}$, and so $Y = C \cup N(C)$ does not cover $\mathcal{B}$. The tree decomposition of H_C with bags Y and X is $\mathcal{B}$-admissible, and we are done.

So we may assume that $\mathcal{B}'$ is a bramble of G. As C is disjoint from the cover X of $\mathcal{B}$, it is not in $\mathcal{B}$, so $\mathcal{B}'$ is a larger bramble than $\mathcal{B}$. Thus by the induction hypothesis, G has a $\mathcal{B}'$-admissible tree decomposition (T, V). If (T, V) it is also $\mathcal{B}$-admissible, we are done, so we may assume that some bag, V_s, of size greater than k, covers $\mathcal{B}$.

Since $\ell = |X|$ is the size of the smallest cover of $\mathcal{B}$, we cannot disconnect the cover X of $\mathcal{B}$ from the cover V_s of $\mathcal{B}$ by removing less than ℓ vertices, so by Menger's theorem there is a collection of $\{P_x \mid x \in X\}$ of disjoint paths linking X into V_s.

If V_s contained anything in C, then it would cover $\mathcal{B}'$, contradicting the fact that (V, T) is $\mathcal{B}'$-admissible, so V_s is contained in $G - C$, and so the the paths P_x meet H_C only in X.

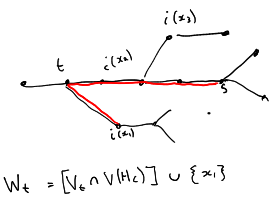


For every element x of X let i_x be a vertex of T such that V_{i_x} contains x. Using (T, V) we make a tree decomposition (T, W) of H_C as follows. For every

bag V_t of (T, V) let

$$W_t = [V_t \cap V(H_C)] \cup \{x \mid t \text{ is in the path in } T \text{ between } i_x \text{ and } s\}.$$

This is a tree decomposition as the elements x that are added to bags are added coherently– they are added to V_t for all t between i_x and s in T.



As $V_s \subset G - C$ we have that $V_s \cap V(H_C)$, and so $W_s = X$ as s is in the path from i_x to s in T for all $x \in X$.

Now, for any t in T, we have that $|W_t| \leq |V_t|$. Indeed for any x added into W_t we claim that there is a vertex of P_x in V_t that is not in H_C. This is because P_x is a path with one endpoint in V_{i_x} and another in V_s, so it must have an element in V_t because t is between i_x and s in T.

The last thing to check is that (T, W) is a $\mathcal{B}$-admissible decomposition of H_C. So consider a bag W_t of size greater than k. This is a bag in H_C and is bigger than X so contains a vertices of C. The original bag V_t then has size at least k and also contains a vertex of C. As (T, V) is $\mathcal{B}'$-admissible, it must then miss some bramble-set $B \in \mathcal{B}$. We claim that W_t misses B also. If not, there is some x in W_t that hits B. But t is in the path in T between i_x and s, and B is hit by both V_{i_x} and V_s, but is not hit by V_t. This is a contradiction. $\square$

2.3.1 Tree width and chordal graphs and cliques

There are a couple of other useful definitions of tree width. Let (T, V) be a tree decomposition of a graph G with width k. So we have bags V_i of size at most $k+1$ for each vertex of T, and the indices of the bags containing a given vertex g of G induce a subtree T_g of G. Let's now further throw in edges so that each bag is complete. One can show that the graph G' that we get in this way is chordal, and that the bags are its maximal cliques. The tree decomposition becomes the intersection graph of the subtrees T_g of T. From this we get the following.

Fact 2.15. *A graph G is chordal if and only if it has a tree decomposition in when every bag is complete.*

And the following.

Fact 2.16. *The tree width of a graph G is one less than the minimum of the clique numbers of the chordal graphs that contain it as a subgraph.*

It is well known that a chordal graph with clique number $k+1$ can be constructed from a clique (of size at most $k+1$) by adding new vertices adjacent to all vertices of a clique (of size at most k). A graph of tree width k is a subgraph of such a graph, so we have the following.

Fact 2.17. *A graph of tree width k can be constructed from a bag of size k by adding vertices adjacent to at most $k-1$ vertices of a bag.*

It doesn't have to be done bag by bag. A k-clique sum of two graphs G and G' identifies a k-clique of G with one of G' and then possibly removes edges of the clique. One can show.

Fact 2.18. *A graph has tree width k if and only if it has size at most $k+1$, or is a k-clique sum of two graphs of tree with at most k.*

2.4 Tree width and Forbidden Minors

We observed that a graph property, that is, a family of graphs closed under isomorphism, which is closed under minors, can be decribed as $\mathrm{Forb}_{\preccurlyeq}(\mathcal{M})$ for its minimal family $\mathcal{M}$ of forbidden minors. We also observe that tree-width is closed under taking minors. So there is a finite family $\mathcal{M}$ such that the class of graphs of tree-width at most 2 can be described as $\mathrm{Forb}_{\preccurlyeq}(\mathcal{M})$. So what is that family.

We know that if a graph has a K_4-minor, then it has tree at least 3.

The proof of the following theorem is littered through-out Diestal's book. (The proof references several earlier results. Probably they are not all necessary. Can anyone find a short stand alone proof?)

Theorem 2.19. *A graph has tree width less than 3 if and only if it has no K_4 minor.*

Tree width 2 is clear. One then asks, can we characterise the graphs of tree width less than k, for higher k? We know that the family of forbidden minors is finite, after all. But the family of minors we must exclude grows quickly. One might also ask for what other H does $\mathrm{Forb}_{\preccurlyeq}(H)$ have bounded tree width?

We can say something about this. Such an H must be planar. Indeed all grids (and their minors) are planar, so we cannot omit them without omitting some planar minor; and they have unbounded tree width. So to bound tree width, we must omit a planar minor.

The converse is quite deep.

Theorem 2.20 (Robertson, Seymour '86)**.** *The class* $\mathrm{Forb}_{\preccurlyeq}(\mathcal{M})$ *has bounded tree width if and only if* $\mathcal{M}$ *contains a planar graph.*

Before we discuss the proof of this, (we will at most sketch it), we look at an application.

2.4.1 The Erdős-Póza Property

Recall the Erdős-Póza Theorem.

A *(cycle) kiling set* of a graph G is a subset X of vertices of G such that every cycle in G contains at least one vertex in X. (Removing the set X 'kills' all cycles.) Clearly no cycle killing set can have fewer vertices than the number of disjoint cycles in G. If k is the maximum number if disjoint cycles in a graph is there necessarily a killing set of size k? (No; K_4 is a counter-example.)

Theorem 2.21. *There is a function* $f : \mathbb{N} \to \mathbb{N}$ *such that for any* $k \geq 1$*, every graph has at least* k *disjoint cycles, or a cycle killing set of size at most* $f(k)$*.*

Generalising this, a family $\mathcal{H}$ of graphs has the *EP-property* if there is a function f such that any graph has more than k disjoint copies of graphs in $\mathcal{H}$, or an $\mathcal{H}$ killing set of size at most $f(k)$.

It is clear that $\mathcal{H} = \{H\}$ for any one graph H has the EP-property with $f(k) = |V(H)| \cdot k$. But cycles can be arbitrarily big, so it is not trivial that cycles have the EP-property.

Now, any cycle has a K_3-minor, and any graph with a K_3-minor contains a cycle. So killing all cycles is the same as killing all K_3-minors. Thus Erdős and Pósa show that the class $\mathcal{M}_{K_3}$ of graphs with a K_3-minor has the EP-property. What other minor defined classes $\mathcal{M}$ have the EP-property?

Corollary 2.22. *If* H *is planar, then the class* $\mathcal{M}_H$ *of graphs with* H *as a minor has the EP-property.*

Proof. The proof is an induction on k where $f(0) = 0$ and $f(k) = 2 \cdot f(k-1) + w_k$ where w_k is the tree width from Theorem 2.20 that implies the existence of the graph consisting of k disjoint copies of H (which is planar as H is).

If a graph G has tree width at least w_k then we are done, so we may assume it has a tree decomposition (T, V) of width less than w_k.

Consider the case that there is some edge ij of T such that U_i and U_j both contain copies of H as a minor. If either of them contains more than $k-1$ disjoint copies, we have more than k copies in G and we are done. If together they have fewer than k disjoint copies, then they each at most $k-1$ disjoint copies, and so by induction we can kill all copies in G by removing at most $f(k) = 2f(k-1) + w_k$ vertices, and so are done.

So for any edge ij of T at most one side U_i or U_j contains a graph with H as a minor, so we can direct the edge. The same trick as before allows us to find a bag whose removal kills all H-minors. $\square$

2.4.2 On the proof of Theorem 2.20

We have to show that if we omit some planar minor in a class, then the class has bounded tree width. We show that if a class has unbounded tree width, then all planar graph are minors of the graphs of the class. It is not too hard to see that every planar graph is a minor of some grid. Just draw the graph on fine enough graph paper and remove all vertices not covered by your think pencil lines. So to prove Theorem 2.20 one only has to show.

Theorem 2.23. *For every integer $r \geq 1$ there is an integer k such that every graph of tree width at least k contains an $r \times r$-grid minor.*

As K_{r^2} contains a $(r \times r)$-grid minor, this can be proved by proving the following.

Theorem 2.24. *For all m and r, there is an integer k such that every graph of tree width at least k has a K_m minor, or an $r \times r$-grid minor.*

The proof of this takes almost 10 pages in Deistal. It is worth trying to read, but I don't think I can cover it properly here.

In very broad strokes, the main attack is to find a bunch of disjoint connected vertex sets $A_1, \ldots, A_m$ in G such that each pair i, j has a big linkage $P_{i,j}$: sets of many disjoint A_iA_j-paths. If for some i, j, i', j' these linkages $P_{i,j}$ intersect too much, then one can find (with a lot of work) a $r \times r$-grid minor. If not, then one can find a representative path from each linkage such that representative paths are disjoint. This gives a K_m minor.

2.4.3 The Graph Minor Structure Theorem

We stated above that if we omit some planar minor, then our class has bounded tree width. What happens if we omit other, non-planar graphs, as minors. The Graph Minor Structure Theorem says that in this case, and graph in the class has a tree decomposition, in which bags (cannot be assumed to have bounded size, but) are nice is the sense that they are 'almost planar' and the non-planar bits are well controlled by being in a low width path decomposition.

(The important part is that the bags are controlled enough that they can be showed to be well ordered under the minor order, so that trees of them can be well ordered to get the Graph Minor Theorem.)

We will state the Graph Minor Structure Theorem for cliques– giving the structure of graphs that omit K_n as a minor. This requires some definitions.

Given a tree decomposition (T, V) of a graph G, the *torsos* of G are the graphs H_i that we get from the subgraphs $G[V_i]$ induced by the bags V_i by completing to a clique any intersection $V_i \cap V_j$ with another bag.

It is easy to see that G can then be constructed by clique-sums from its torsos, so gets various bounds from those on its torso:

i. $\chi(G) \leq \max \chi(H_i)$, and

ii. $\omega(G) \leq \max \omega(H_i)$, for example.

A graph H has a *k-near embedding* in a surface S if there is a set X of at most k vertices such that there is a decomposition

$$H \setminus X = H_0 \cup H_1 \cup \cdots \cup H_k$$

such that H_0 is embedded on the surface S' that we get from S by removing k disjoint disks, and for each i the graph H_i has an embedding on the ith removed disk such that its boundary– those vertices with neighbours outside H_i, are embedded on the boundary of the disk (so defining a cycle C on the vertices), and H_i has a cycle decomposition (C, V) of width at most k, where for each vertex of c, the vertex c of H_i is in the bag V_C of the decomposition.

PUT a picture HERE.

The Graph Minor Structure Theorem for cliques is as follows.

Theorem 2.25 (Roberson, Seymour '03)**.** *For every $n \geq 5$ there is an integer k such that any graph G in $\mathrm{Forb}_{\preccurlyeq}(K_n)$ has a tree decomposition whose torsos are k-nearly embedded into a surface in which K_n cannot embed.*

Note that insisting the torsos embed into a surface into which K_n cannot embed bounds their genus, so stops the torsos from being too far from planar. This theorm is used in the proof of the Graph Minor Theorem, but apparently, has many applications apart from this. (Hopefully we will get to some.)

2.5 The Graph Minor Theorem

The graph minor theorem says that any minor closed class is defined by finitely many omitted minors. As a special case of this, the set of graph embeddable on any surface S is $\mathrm{Forb}_{\preccurlyeq} \mathcal{M}_S$ for some finite family $\mathcal{M}_S$ of minors. Showing this special case is easier than showing the Graph Minor Theorem in general, and (apparently) gives a good feel for many of the ideas in the proof of the Graph Minor Theorem.

To show that $\mathrm{Forb}_{\preccurlyeq} \mathcal{M}_S$ is finite for any surface S one argues that the set $\mathcal{M}_S$ must have unbounded tree-width, or we are done by the fact that any family of bounded tree width is well ordered. So for all r, some graph M in

$\mathcal{M}_s$ has a $(r \times r)$-grid minor. The following lemma then gives us our proof by contradiction.

Lemma 2.26. *For every surface S there is some r such that no graph that is minimally non-embeddable in S contains the grid H^r as a topological minor.*

The grid H^r is the hexagonal grid of radius r. It is easy to show that the families $\{H^r \mid r \in \mathbb{Z}\}$ and $\{G_{n\times n} \mid n \in \mathbb{Z}\}$ are equivalent in the sense that every graph in one family is a minor of a graph in the other family. The family of hexagonal grids is nice because it has max degree 3, and it is well known that a graph of max-degree 3 is a minor of a graph if and only if it is a topological minor. Recall that M is a topological minor of a graph G if there is a subdivision of M that is a subgraph of G.

This lemma though is also quite long– 4-5 pages in Diestal. I am having a bit of trouble understanding it.

3 From Dvořak's Text

We now switch to Zdeněk Dvořak's new book "Graph Minor: Theory and Applications". We will use it to fill in a bit more theory, but will hopefully get quickly to some applications. We change notation a little to stay consistant with Dvořak's notation.

An H-minor in a graph H is now a *model* μ mapping vertices of H to subsets of vertices of G. As one is more often interested in finding a path between two particular vertices of a graph G than just in finding a path in G, one will often want to find a minor in which certain vertices of G are in pre-assigned branch sets. Where ρ is a function that assigns vertices of H to some of the vertices of G, a *ρ-rooted*-minor μ of H in G is a minor in which a vertex v of G is in the branch set $\mu(\rho(v))$ if $\rho(v)$ is defined.

A tree decomposition of a graph G is now usually denoted by (T, β). Vertices of T are called *nodes*, to differentiate them from the vertices of G which are called vertices.

We start at 2.4 of Dvořak.

3.2.4 Balanced Separators and Unbreakable sets

We will define an obstruction to tree width $< k$ called a k-unbreakable set– a set that is 'almost k-connected' in the sense that by removing fewer than k-vertices, we can only break off 'small corners', leaving some big component that contains most of the set. A $k \times k$-grid will be an example of a k-unbreakable set.

To talk about k-unbreakable, we need a notion of 'breaking'- this comes in the form of balanced separations.

Recall that a *vertex cut* S of a connected graph G is set of vertices for which $G \setminus S$ is disconnected. Partitioning the components of $G \setminus S$ into two non-empty subgraphs A' and B', the subgraphs A and B induced by $V(A') \cup S$ and $V(B') \cup S$, are a *separation* (A, B) of G. Both a vertex cut, or a separation, are called balanced if none of the 'parts' they leave are too big. For the precise definition it is useful to give a more general version using a vertex weighting.

For an assignment $\omega : V(G) \to \mathbb{R}^+$ of non-negative weights to the vertices of G, the weight on any subgraph H, (or $V(H)$), is the sum

$$\omega(H) := \sum_{v \in V(H)} \omega(v)$$

of the weights of its vertices. A vertex cut S of a graph G with weight ω is *ω-balanced* if for each component C of $G \setminus C$ has $\omega(C) \leq \frac{2}{3}\omega(G)$. A separation (A, B) of G is ω-balanced if

$$\omega(A \setminus B) \leq \frac{2}{3}\omega(G) \qquad \text{and} \qquad \omega(B \setminus A) \leq \frac{2}{3}\omega(G).$$

The acual constant $2/3$ is not so important, what is important is that there is some constant. In an recursive process where we find a cutset and break the graph into parts, this bounds the number of times we have to iterate by a logarithm, so leads to a polynomial algorithm. Dvořak likes to use $2/3$ though because (one can show as a simple exercise that reinforces the relation of cutsets to separations) that this is the value at which we get that a weighted graph G has a ω-balanced separation if and only if it has a ω-balanced cut-set.

For a tree T, orienting each edge e towards the heavier component of $T \setminus e$, we get a sink t whose removal leaves no component of weight more than $(1/2)\omega(T)$, so every tree has a balanced cutset of size 1. For a tree decomposition (T, β) of graph G with weighting ω_G, we get a weighting ω_T of T by setting $\omega_T(t)$ for a node t to be $\omega_G(\beta(t))$. Only, for vertices v of G in more than one bag $\beta(t)$, we split the weight among the nodes t for which $v \in \beta(t)$; so that $\omega_T(T) = \omega_G(G)$. The ω_T-balanced cutvertex t of T yields an ω_G-balanced cut $\beta(t)$ of G giving the following.

Lemma 3.1. *For any weighting ω of a graph G of tree-width k there is an ω-balanced cut of G of size at most $k + 1$.*

The most natural weighting ω of a graph G is the weighting that assigns a weight of 1 to each vertex. With this we get that graph of tree-width k has a cut of size at most $k + 1$ that breaks G into components none of which contain more than $2/3$ of the vertices of G.

But one can also take ω to just assign a weight of one to vertices in some subset X of the vertices of G, and 0 to the other vertices. If G has tree-width k then there is a cut of size at most $k + 1$ such that breaks this set up (in a

balanced way) among the components. Finding a set that cannot be broken then is a certification of large tree width.

A cut (or separation) of G is X-balanced if it is ω_X-balanced where ω_X is the characteristic function of X for a subset X of $V(G)$. A set X is *k-unbreakable* if there is no X-balanced cut of size less than k. (Notice that a k-unbreakable set X must have size at greater than $3k-3$, as if we remove ane $k-1$ vertices, there must be a component containing at least $2k-2$ vertices of X.)

Lemma 3.2. *If G has a tree-width at least $4k-3$ then it has a k-unbreakable set of size $3k-2$.*

Proof. Assuming that G has no such k-unbreakable set, so for every set R of size $3k-2$ there is an R-balanced cut S of size at most k, we find a tree decomposition of G size less than $4k-3$. This is by induction on $|V(G)|$ on the slightly stronger statement that for every induced subgraph G' of G and every subset $R \subset V(G)$ of size $3k-2$, there is a tree decomposition of G' of tree-width less than $4k-3$ of which R is a bag.

This clearly holds when $|V(G)| \leq 4k-3$.

Let G' and the subset R of G be given, so there is an r-balanced cut S of size at most k. This a separation (A, B) with $A \cap B = S$. As $A \cap R$ has size at most $2k-2$ even adding all of S, $R_A := (A \cap R) \cup S$ has size at most $3k-2$, and we can add vertices of A to bring it up to size $3k-2$. By induction there is a tree decomposition of A of width at most $4k-3$ with R_A in a bag. Similarily there is a tree decomposition of B of width at most $4k-3$ with R_B in a bag. Joining these by adding a bag X containing $R \cup S$ (which has size at most $4k-3$) adjacent (in the new tree) to the bags containg R_A and R_B, we get the tree decomposition of G that we needed. $\square$

3.2.5 Tangles

We saw that if a graph has tree width $4k$, then it has a k-unbreakable set. Here we show that if a graph has a k-unbreakable set, then it has tree width at least k. We do so via the notion of a tangle, which will turn out to be a very useful obstruction to small tree width, if a little abstract.

A *system $\mathcal{T}$ of $(< k)$-separations* for a graph G is an orientation of the separations of G of order less than k. That is, exactly one of (A, B) and (B, A) is in $\mathcal{T}$ for every pair of subgraphs A, B of G that are a separation of order less than k.

(We will actually only care about the vertex sets of A and B, an so will consider what are called systems of $(< k)$-vertex-separations, the book deals with this version separately, but for our purposes, for now, the vertex version is enough.)

Given a k-unbreakable set X of a graph G, we have for any $(< k)$-separation (A, B) of G that

$$|X \setminus A| \geq \frac{2}{3}|X| \qquad \text{or} \qquad |X \setminus B| \geq \frac{2}{3}|X|.$$

Make a system $\mathcal{T}_X$ of $(< k)$-separations of G by putting (A, B) in $\mathcal{T}_X$ if $|X \setminus B| \geq (2/3)|X|$; it becomes a digraph where edges are directed towards the X-heavy part of a separation.

This system has the following property.

$(T1)$: If $(A_i, B_i) \in \mathcal{T}_X$ for $i \in \{1, 2, 3\}$ then $G \neq A_1 \cup A_2 \cup A_3$.

A system of $(< k)$-separations satisfying $(T1)$ is a *tangle* of order k in G.

Fact 3.3. *If G has a k-unbreakable set X then it has a tangle $\mathcal{T}_X$ of order k.*

We will show that a tangle of order k gives tree width at least k. This will use some simple observations about tangles.

Fact 3.4. *Let $\mathcal{T}$ be a tangle of order k in G.*

i. If $(A, B) \in \mathcal{T}$ and (A', B') is a $(< k)$-separation of G with $A' \subset A$, then $(A', B') \in \mathcal{T}$.

ii. If $(A_i, B_i) \in \mathcal{T}$ for $i \in \{1, \ldots, n\}$ and $C = \bigcup_{i=1}^{n} A_i \cap B_i$ has size less than k, then $(A, B) = (\cup A_i, \cap B_i)$ is a $(< k)$-separation and is in $\mathcal{T}$.

Proof. Part i) is trivial. Part ii) is trivial if $n = 1$, and we prove it by induction on n. Indeed for $n \geq 2$ let $(A', B') = (\cup_{i=1}^{n-1} A_i, \cap_{i=1}^{n-1} B_i)$, this is in $\mathcal{T}$ by the induction hypothesis; as is (A_1, B_1). As $(A' \cup A_n \cup B) = G$, we have by property $(T1)$ that (B, A) is not in $\mathcal{T}$, and so (A, B) is. □

Theorem 3.5. *If G has a tangle of order k, then* $\mathrm{tw}(G) \geq k - 1$.

Proof. Assume, towards contradiction, that some graph G with a tangle $\mathcal{T}$ of order k has a tree decomposition (T, β) of width less that $k - 1$. Each edge e of T yields a $(< k)$-separation into subgraphs $\bigcup T_a$ and $\bigcup T_b$, and so $\mathcal{T}$ assigns an orientaiton of e towards the bigger one. This orientation yields a sink t in T and its bag $\beta(t)$. Each edge e into t gives a $(< k)$-separation (A_e, B_e) in $\mathcal{T}$, and the $< k)$-separation $(\beta(t), G)$ is also in $\mathcal{T}$. Since each $A_e \cap B_e$ is in $\beta(t)$, we can apply part ii) of the fact to get that

$$(\beta(t) \cup \bigcup_e A_e, \bigcap_e B_e) \in \mathcal{T}.$$

But as $\beta(t) \cup \bigcup_e A_e = V(G)$, this is impossible by property $(T1)$. □

Corollary 3.6. *If G has a k-unbreakable set, then* $\mathrm{tw}(G) \geq k-1$.

Now, we showed that if $\mathrm{tw}(G) \geq k-1$ then G has a $k/4$-unbreakable set, so has a tangle of order $k/4$. Roberston and Seymour improve this to show it has a tangle of order $(2/3)k$.

3.2.6 k-unbreakable sets to brambles

Though we get this from the previous section using the fact that G has a bramble of order k if and only if it has tree width at least $k-1$, we can show it directly.

Fact 3.7. *If G has a k-unbreakable set then it has a bramble of order k.*

Proof. Let X be a k-unbreakable set in G, and from it define the bramble whose bramble sets are all sets B inducing connected subgraphs of G that contain more than half the vertices of X. This is indeed a bramble, as any two sets share a vertex of X. Let H be a hitting set of the bramble (what Diestal called a cover– it contains a vertex from every brambles set B). When we remove it from G we kill all connected subgraphs that contain at least half the vertices of X, and so leave a graph non component of which contains more than half of X. So H is an X-balanced cut of G. As X is k-unbreakable, it has size at least k, so the bramble has order at least k.

□

3.2.8 Comparison of Obstructions

We have seen essential several obstructions to small tree width– graded structures such that a graph has large tree width if and only if it has a large such structure: unbreakable sets, brambles, tangles, grid-minors and wall-minors. Dvořak compares them in this section but finishes off by saying that tangles tend to be used most often, though they 'give a little tree width away'. The grid (wall) minors are also extremely usable, but give away a lot of tree width– when we use them we usually aren't looking for something tight in tree width, only that it exists.